RFMiD Retinal Disease Classification using CNN Models

Student: Manaswi Patil
Course: Summer Training – Week 2
Lab: VISLab

---

1. Introduction

Retinal disease detection is an important application of computer vision in medical imaging. In this project, convolutional neural networks (CNNs) were trained on the RFMiD (Retinal Fundus Multi-Disease Image Dataset) to classify retinal disease images.

The objectives of this experiment were:

- To compare the performance of different CNN architectures.
- To analyze the effect of Focal Loss on class imbalance.
- To monitor training behavior using Weights & Biases (wandb).

Four experiments were conducted using different model architectures and loss functions.

---

2. Dataset

The dataset used in this project is the RFMiD (Retinal Fundus Multi-Disease Image Dataset).

Dataset characteristics:

Property	Value
Dataset	RFMiD
Images	~3200 retinal images
Classes	28 retinal diseases
Image Size	Resized to 224 × 224

Each sample consists of a retinal fundus image and its corresponding disease label.

A custom PyTorch dataset loader (rfmid_dataset.py) was implemented to load images and labels for training.

3. Training Configuration

All experiments were trained with the following settings:

Parameter	Value
Batch Size	32
Learning Rate	1e-4
Epochs	10
Optimizer	Adam
Image Size	224 × 224
GPU	Tesla V100

Training progress was tracked using Weights & Biases.

---

4. Experiments

The following models and configurations were evaluated:

Experiment	Model	Loss Function
Run 1	ResNet18	CrossEntropyLoss
Run 2	ResNet18	FocalLoss
Run 3	ResNet50	CrossEntropyLoss
Run 4	VGG16	CrossEntropyLoss

These experiments allow comparison of:

- model complexity
- training convergence
- classification performance

5. Training Loss Analysis

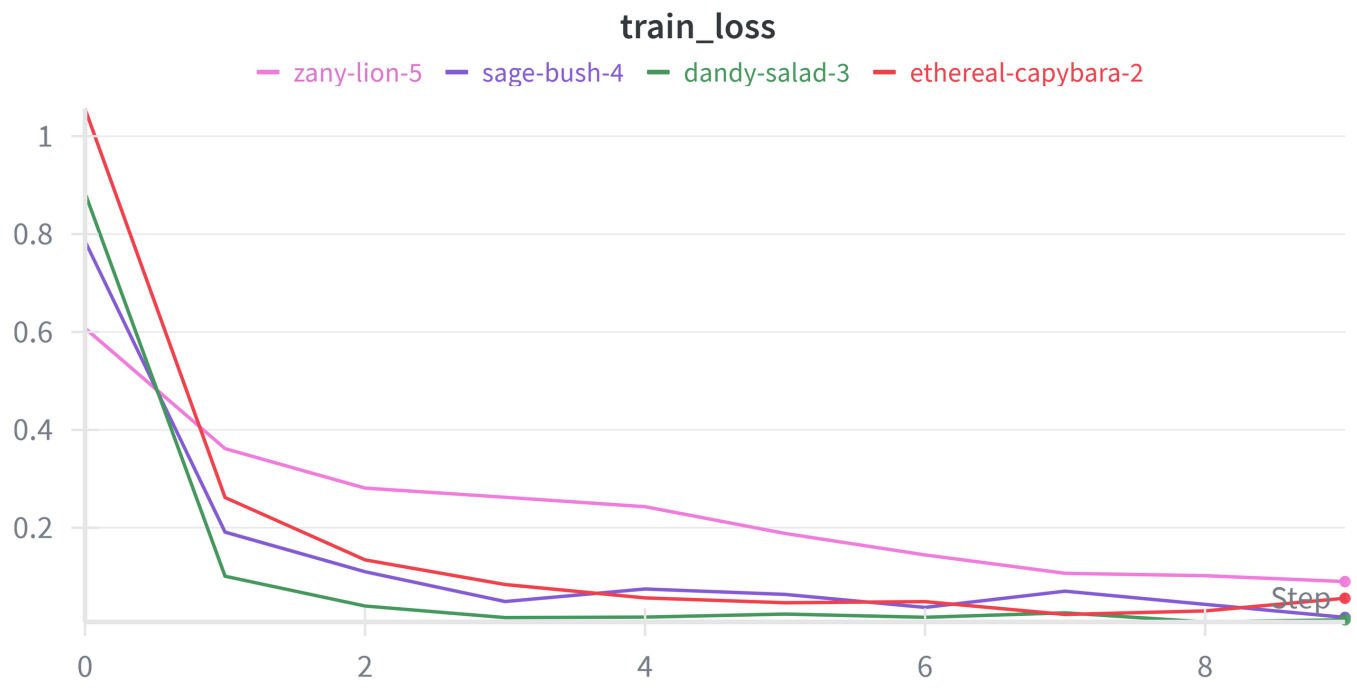

Figure 1: Training loss comparison across different CNN models during RFMiD dataset training.

The training loss curves show that all models converge rapidly during the first few epochs, indicating that the networks quickly learn meaningful features from the retinal images.

The **ResNet18 model trained with Focal Loss (green)** demonstrates the fastest convergence and achieves the lowest training loss throughout most epochs.

The **ResNet18 model trained with CrossEntropyLoss (red)** starts with the highest loss but quickly stabilizes after the first epoch.

The **ResNet50 model (purple)** also shows strong convergence but with slightly more fluctuation compared to ResNet18.

The **VGG16 model (pink)** decreases more gradually and maintains the highest loss values, suggesting slower optimization compared to the ResNet-based architectures.

Interestingly, although the ResNet18 model trained with Focal Loss achieved the lowest training loss, its validation accuracy is similar to the CrossEntropyLoss version. This suggests that lower training loss does not always directly translate to higher validation accuracy, highlighting the importance of evaluating models using validation metrics rather than training loss alone.

6. Validation Accuracy Analysis

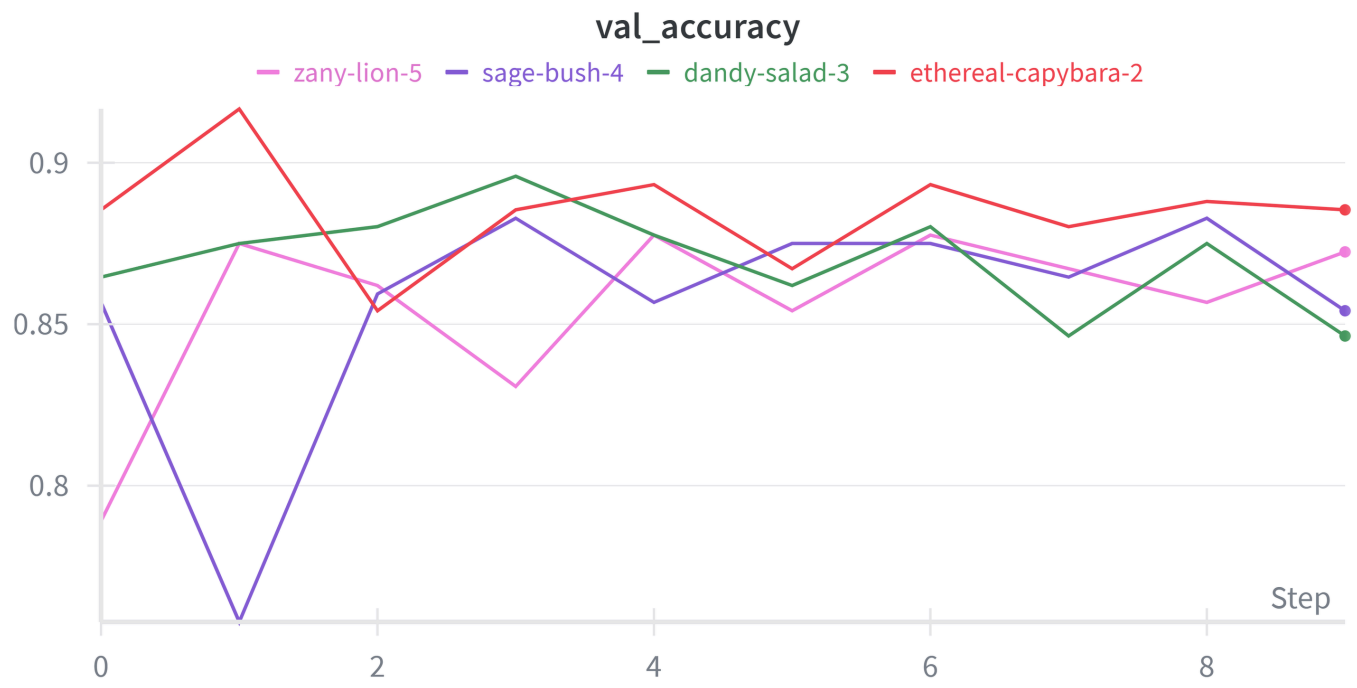

Figure 2: Validation accuracy comparison across different CNN models during RFMiD dataset training.

The validation accuracy curves show that all models achieve strong performance, with accuracies ranging approximately between **0.85 and 0.92**.

The **ResNet18 model trained with CrossEntropyLoss (red)** achieves the highest peak validation accuracy during training, indicating strong generalization performance.

The **ResNet18 model trained with Focal Loss (green)** maintains relatively stable validation accuracy across epochs, suggesting that Focal Loss helps maintain consistent performance.

The **ResNet50 model (purple)** shows slightly larger fluctuations early in training but stabilizes in later epochs.

The **VGG16 model (pink)** shows slower improvement and slightly lower overall accuracy compared to the ResNet-based architectures.

While training loss consistently decreases across epochs, validation accuracy fluctuates slightly. This suggests that the models may experience minor overfitting effects, which is common when training deep models on relatively small medical datasets.

7. Effect of Focal Loss

Focal Loss was applied to the ResNet18 model to address class imbalance in the RFMiD dataset. Compared with standard CrossEntropyLoss, the Focal Loss configuration demonstrated faster training convergence and maintained stable validation accuracy. This suggests that emphasizing harder training samples may help the model learn more robust representations for imbalanced medical datasets.

---

8. Conclusion

This study demonstrates that CNN architectures are effective for retinal disease classification using the RFMiD dataset. Among the tested models, **ResNet18 achieved strong performance with efficient convergence**, while the **Focal Loss configuration provided faster training convergence and stable validation performance**. Deeper architectures such as ResNet50 also performed well but did not significantly outperform the simpler ResNet18 model. These results highlight the effectiveness of transfer learning and modern CNN architectures for medical image classification tasks.